

Java toString() Method

If you want to represent any object as a string, **toString() method** comes into existence.

The toString() method returns the String representation of the object.

If you print any object, Java compiler internally invokes the toString() method on the object. So overriding the toString() method, returns the desired output, it can be the state of an object etc. depending on your implementation.

Advantage of Java toString() method

By overriding the toString() method of the Object class, we can return values of the object, so we don't need to write much code.

Understanding problem without toString() method

Let's see the simple code that prints reference.

Student.java

```
class Student{
    int rollNo;
    String name;
    String city;

    Student(int rollNo, String name, String city){
        this.rollNo=rollNo;
        this.name=name;
        this.city=city;
    }

    public static void main(String args[]){
        Student s1=new Student(101,"Raj","lucknow");
        Student s2=new Student(102,"Vijay","ghaziabad");
    }
}
```

```
System.out.println(s1);//compiler writes here s1.toString()
System.out.println(s2);//compiler writes here s2.toString()
}
}
```

Output:

```
Student@1fee6fc
Student@1eed786
```

As you can see in the above example, printing s1 and s2 prints the hashcode values of the objects but I want to print the values of these objects. Since Java compiler internally calls toString() method, overriding this method will return the specified values. Let's understand it with the example given below:

Example of Java toString() method

Let's see an example of toString() method.

Student.java

```
class Student{
    int rollNo;
    String name;
    String city;

    Student(int rollNo, String name, String city){
        this.rollNo=rollNo;
        this.name=name;
        this.city=city;
    }

    public String toString(){//overriding the toString() method
        return rollNo+" "+name+" "+city;
    }

    public static void main(String args[]){
        Student s1=new Student(101,"Raj","lucknow");
        Student s2=new Student(102,"Vijay","ghaziabad");
    }
}
```

```
System.out.println(s1);//compiler writes here s1.toString()
System.out.println(s2);//compiler writes here s2.toString()
}
}
```

Output:

```
101 Raj lucknow
102 Vijay ghaziabad
```

In the above program, Java compiler internally calls **toString()** method, overriding this method will return the specified values of **s1** and **s2** objects of Student class.

Java toString() method MCQ

1. What is the default behavior of the toString() method in the Object class?

1. Returns the object's state
2. Returns the object's class name and hashCode
3. Returns null
4. Returns the object's memory address

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2. Which class should override the toString() method to provide a meaningful string representation?

1. Any class that requires a custom string representation
2. Only utility classes
3. Only data classes
4. Only service classes

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3. What is the recommended way to generate a string representation of an object in a custom toString() method?

1. Use manual string concatenation
2. Use `String.format()` or `StringBuilder` for better performance
3. Use the default implementation
4. Avoid overriding the method

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4. What will be the output of the `toString()` method if it is not overridden in a subclass?

1. The class name and hashCode
2. Null
3. The object's state
4. The object's memory address

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5. In which package is the `Object` class, which defines the `toString()` method, located?

1. `java.lang`
2. `java.util`
3. `java.io`
4. `java.net`

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